Oluwatosin Ayeni

Senior ML Engineer (Token Factory) — Nebius

Glasgow, Scotland, UK · 07727305230 · ayenioluwatosinolawale@gmail.com
github.com/AyeniOluwatosinOlawale · linkedin.com/in/oluwatosinolawaleayeni

PROFILE

Senior ML Engineer specialising in LLM inference serving, speculative decoding, and production model deployment. Benchmarked vLLM and SGLang on 2× NVIDIA H100 NVL 94 GB — 1,443× TTFT speedup via RadixAttention prefix caching, 5,965 tok/s peak. Delivered 28× cost reduction via continuous batching and KV cache optimisation. Experienced across the full model lifecycle: pre-training, LoRA/PEFT fine-tuning, AWQ/GPTQ quantisation, speculative decoding, and production serving. Open to London hybrid or UK remote.

CORE COMPETENCIES

LLM Inference & Serving	vLLM, SGLang, TensorRT-LLM, LiteLLM Proxy; speculative decoding, continuous batching, KV cache optimisation, PagedAttention, RadixAttention, prefix caching
Model Efficiency	AWQ/GPTQ/BnB quantisation, LoRA/PEFT fine-tuning, low-precision inference (BF16, INT8, INT4), model compression, throughput/latency Pareto analysis
GPU & CUDA Optimisation	NVIDIA Nsight Systems/Compute, CUDA kernel profiling, memory bandwidth analysis, tensor parallelism (TP=2 on H100 NVLink), multi-GPU inference
Training & Fine-tuning	Mistral 7B, Phi-4, Qwen3, Gemma 2 via LoRA/PEFT; complete training pipelines; trained LM from scratch (Qwen3-TinyStories)
Infrastructure & Serving	Kubernetes (GKE), KEDA, Argo CD, Prometheus, Grafana, Docker, FastAPI, Python, Terraform, Redis

PROFESSIONAL EXPERIENCE

Jan 2025–present — **Founder & Lead Engineer** — *Nexora Academic*, Glasgow, UK

- H100 NVL benchmark: vLLM v0.30 vs SGLang v0.5.20 — Qwen3-8B BF16, TP=2. Peak: 5,965 tok/s; RadixAttention: 1,443× TTFT speedup (39 ms vs 55,565 ms) at 131K-token context; 100% SLO compliance up to concurrency 128. Live: inference-web-theta.vercel.app
- Built *inference-optimizer*: production AI Inference Gateway — 3.87× faster (parallel execution), 61.4% cost reduction (prompt caching), 14× throughput (concurrency), 64.3% cheaper (model routing). 13 optimisations, 4 providers.
- Built *vllm-k8s-stack*: Kubernetes LLM serving — KV-cache-aware routing, HPA autoscaling on GPU pools, Prometheus/Grafana observability.
- Built *inference-harness*: unified benchmark framework spanning closed APIs + vLLM/SGLang/TensorRT-LLM with GPU profiling and live dashboard.
- Designed 13-phase AI Inference Engineering curriculum covering GPU architecture, CUDA optimisation, serving frameworks, and distributed inference.

Jan 2023–present — **Founder & AI Consultant** — *Vertinab / AnaburyAI Technology*, Glasgow, UK

- 28× LLM cost reduction deploying Qwen2.5-7B on RTX 4090 via vLLM/SGLang with continuous batching and KV cache optimisation.
- CUDA KV cache kernel profiling with NVIDIA Nsight: identified memory bandwidth bottleneck; 3.8× throughput improvement; 18× latency reduction via continuous batching redesign.
- Fine-tuned Mistral 7B with LoRA/PEFT on radiology corpus: +22% diagnostic accuracy, −35% hallucination rate; deployed as production clinical inference API.

2024–present — **Senior Software Engineer (Contract)** — *EduTechConnect / Stellar Labs*, Remote

- Real-time NLP pipeline (WhatsApp → Zoho CRM → Supabase) with sub-500 ms end-to-end latency under concurrent load.

Mar 2026–present **AI Automation Engineer** — *EduTech Global*, Remote

- LLM-backed student chatbots with structured output extraction and multi-step reasoning for Ahmadu Bello and Babcock universities.

2018–2022 **Data Analyst** — *Bytelab Technologies Ltd*, Nigeria

KEY PROJECTS

llm-inference-benchmark (H100) vLLM vs SGLang on 2× H100 NVLink, Qwen3-8B BF16 TP=2. 5,965 tok/s peak; 1,443× TTFT speedup via RadixAttention. Live inference-web-theta.vercel.app

open-model-engine GPU-first benchmarking: vLLM, SGLang, TensorRT-LLM, speculative decoding, quantisation (AWQ/GPTQ), throughput/latency Pareto. GitHub

Qwen3-TinyStories-Training Complete LM training pipeline from scratch: tokenizer training, architecture, training loop, generation eval. GitHub

My_finetuned_Phi4 Phi-4 LoRA fine-tuning: data prep, LoRA config, training, eval, inference. GitHub

llm-inference-platform (GKE) Production serving: vLLM + SGLang + LiteLLM Proxy on GKE; KEDA autoscaling; Prometheus/Grafana; Argo CD GitOps; 1.25M+ tokens. GitHub

EDUCATION

2022–2023 **MSc Artificial Intelligence and Applications, Distinction** — *University of Strathclyde*, Glasgow, UK

2010–2016 **MTech / BSc Physics** — *Federal University of Technology Akure*, Nigeria

OPEN SOURCE

GitHub 43 public repos — github.com/AyeniOluwatosinOlawale